25901408

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION

March 2026 Fourth SEMESTER

Value Addition Course (VAC) – MG4VACMBG200 - SANITATION MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Part A- Short Notes Questions - Answer any 10 questions - Each question carries 2 marks

1. Give two reasons why protective clothing is necessary in fermentation units. [U] / [CO1]

a. Prevents contamination of the fermentation process by microorganisms from the worker.

b. Protects workers from exposure to harmful chemicals and biohazards.

2. Define animal inoculation. [K] / [CO1]

Animal inoculation is the introduction of a microorganism or infected material into a susceptible laboratory animal for isolation, identification, or studying its pathogenicity.

3. What is a HEPA filter? [K] / [CO1]

High Efficiency Particulate Air filter, remove99.97% of particles from air as small as 0.3 micrometre.

4. Define droplet nuclei? [U] / [CO2]

Very small in size(less than 5 micrometres), they remain suspended in the air for a long time.

5. Define air sampling devices? [U] / [CO2]

Instruments to collect airborne microbes.

6. What are the preventive measures of diphtheria? [U] / [CO2]

DPT vaccine, isolation of the patient, and maintaining hygiene

vaccination, Managing Exposure, Routine checkups, Hygiene and Environmental Controls

7. Name two common bacteria found in sewage. [K] / [CO3]

Escherichia coli (E. coli), Salmonella typhi, Vibrio cholerae

8. Define primary treatment. [K] / [CO3]

Primary treatment is a physical process designed to remove large suspended solids and grit from wastewater. It involves screening to catch debris, grit removal for inorganic solids, and sedimentation in primary clarifiers, where organic matter settles as sludge and fats/oils are skimmed off the surface.

9. Mention the role of methanogens in anaerobic treatment. [K] / [CO3]

Methanogens are anaerobic archaea that play the final role in anaerobic digestion. They convert intermediate products such as acetate, hydrogen, and carbon dioxide into methane (CH_4) and carbon

dioxide (CO_2). This process stabilises organic matter and helps in biogas production and the reduction of organic pollution in wastewater.

10. Define aeration tanks for water pollution. [K] / [CO3]

An aeration tank is a unit in the activated sludge process of wastewater treatment where air or oxygen is supplied to sewage to promote the growth of aerobic microorganisms. These microorganisms decompose and oxidise organic pollutants, thereby reducing the biochemical oxygen demand (BOD) of the wastewater.

11. Name viral waterborne diseases. [K] / [CO3]

Hepatitis A, Polio, Vibrio cholera.

12. Mention the benefits of composting for soil fertility. [K] / [CO4]

Composting improves soil fertility by adding organic matter and essential nutrients to the soil. It also enhances soil structure, water-holding capacity, and microbial activity, which supports better plant growth and soil health. It also reduces the need for chemical fertilisers and improves long-term soil productivity.

Part B - Short Essay Questions - Answer any 4 questions - Each question carries 5 marks

13. What standard precautions are followed in hospitals and laboratories? [K] / [CO1]

Washing hands before and after contact with patients or specimens. Wearing gloves, masks, gowns, and eye protection when required. Proper disposal of needles and sharp instruments; avoid recapping. Proper cleaning and sterilisation of instruments and surfaces. Segregation and safe disposal of infectious waste. Covering the mouth and nose while coughing or sneezing. Careful collection, transport, and processing of samples. Any 5 [5]

14. How does proper sanitation help in preventing food contamination? [K] / [CO1]

Cleaning utensils, surfaces, and hands reduces bacteria, viruses, and parasites that can spoil food or cause disease. Keeping raw and cooked foods separate and using clean equipment stops germs from spreading. Proper washing of fruits, vegetables, and food-contact surfaces removes pesticides and harmful chemicals.[5]

15. Explain the factors affecting the survival of microorganisms in air? [U] / [CO2]

Explain the factors: temperature, humidity, radiation, air current, and the nature of microbes.

16. Explain the impaction method of air sampling. [U] / [CO2]

Definition- Air is forced to strike agar aap surface. Example: Anderson air sampler, Advantage- quantitative method.

17. Explain why nitrogen and phosphorus need to be removed. [U] / [CO3]

Maintaining the C: N Ratio - Odour Control - Prevention of Nutrient Leaching - Mitigation of Greenhouse Gas Emissions.To carry out the process as environment friendly way

18. Explain shortly why the proper disposal of agricultural waste is necessary. [U] / [CO4]

Introduction [2]. Prevents environmental pollution. Reduces the spread of pests and diseases,

Improves soil health, prevents foul odour and unhygienic conditions, and promotes sustainable agriculture.

[5x4 = 20]

Part-C - Essay Question - Answer any 1 questions - Each question carries 10 marks

19. Discuss the applications and importance of air sanitation techniques in public places, hospitals and industries? [A] / [CO2]

Air sanitation- reduction or elimination of microorganisms present in the air to maintain hygienic and safe environmental conditions.[1] . Importance of Air Sanitation - Maintains a hygienic indoor environment and protects public health. **a**.Hospitals: Air sanitation prevents hospital-acquired infections in operating theatres and ICUs. **b**. Microbiology Laboratories: Maintains sterile conditions for microbial culture and research. **c.** Food Processing Industries: Prevents contamination of food products. **d.** Pharmaceutical Industries: Ensures a sterile environment for drug production. **e**. Public Buildings: Used in airports, schools, and offices to improve air quality. [8]

20. Explain the causes, symptoms, transmission, and preventive measures of any two parasitic waterborne diseases. [U] / [CO3]

a.Cryptosporidiosis, Causative agent: Cryptosporidium parvum (a protozoan parasite). Symptoms: Watery diarrhoea, Abdominal cramps, Nausea and vomiting, Fever and dehydration. Transmission: Infection occurs by ingestion of oocysts through contaminated water or food. It can also spread through the fecal–oral route and contact with infected persons or animals. Preventive measures: Drink safe, properly treated water, maintain good personal hygiene and hand washing, avoid consumption of contaminated food and water, and practice proper sanitation and waste disposal.

b.Giardiasis-Causative agent: Giardia lamblia.Symptoms: Diarrhoea (often greasy or foul-smelling), Abdominal cramps and bloating, nausea, fatigue, and weight loss.Transmission: Spread through the fecal–oral route by ingestion of cysts in contaminated water or food. It can also occur through person-to-person contact or poor hygiene. Preventive measures: Drink safe and properly treated or boiled water, maintain good personal hygiene and regular hand washing, ensure proper sanitation and disposal of human waste, and wash fruits and vegetables properly before consumption.

[10x1 = 10]